In [ ]:
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# EXPERIMENT 5 – DATA VISUALIZATION WITH MATPLOTLIB
PIN NO.: 241UDS0111
## AIM
To visualize and analyze a dataset using line charts, scatter plots, and trend comparisons with
Matplotlib.
```

In [ ]:
```
## Step 1: Import the Required Libraries & Dataset
```

In [4]:
```
import pandas as pd
import matplotlib.pyplot as plt

df_241UDS0111 = pd.read_csv("Experiment_5_241UDS0111_Electric_Vehicle_Charging_Data.csv")
df_241UDS0111.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 17 entries, 0 to 16
Data columns (total 13 columns):
 #   Column                       Non-Null Count  Dtype
---  ------                       --------------  -----
 0   Year                         17 non-null     int64
 1   Charging_Stations            17 non-null     int64
 2   Charging_Demand_GWh          17 non-null     float64
 3   Energy_Supplied_GWh          17 non-null     float64
 4   Charging_Cost_INR            17 non-null     int64
 5   Battery_Mineral_Price_USD    17 non-null     float64
 6   Inflation_Rate               17 non-null     float64
 7   Government_Incentive_Crore   17 non-null     int64
 8   EV_Users_Million             17 non-null     float64
 9   Station_Coverage_Index       17 non-null     float64
 10  Demand_Growth_Percent        17 non-null     float64
 11  Cost_to_Mineral_Price_Ratio  17 non-null     float64
 12  Incentive_per_GWh            17 non-null     float64
dtypes: float64(9), int64(4)
memory usage: 1.9 KB
```

In [5]:
```
df_241UDS0111.head()
```

Out[5]: Columns 1–8 of 14

	Year	Charging_Stations	Charging_Demand_GWh	Energy_Supplied_GWh	Charging_Cost_INR	Battery_Mineral_Price_USD	Inflation_Rate
0	2010	120	0.8	0.6	18	42.0	8.1
1	2011	165	1.1	0.9	18	45.5	7.5
2	2012	230	1.6	1.3	17	48.0	6.9
3	2013	315	2.3	1.9	17	52.0	7.2
4	2014	430	3.1	2.6	16	58.5	6.4

Columns 9–14 of 14 · first column repeated

	Government_Incentive_Crore	EV_Users_Million	Station_Coverage_Index	Demand_Growth_Percent	Cost_to_Mineral_Price_Ratio	Incentive_per_GWh
0	25	0.02	6000.00	0.00	0.4286	31.25
1	35	0.03	5500.00	37.50	0.3956	31.82
2	48	0.05	4600.00	45.45	0.3542	30.00
3	62	0.08	3937.50	43.75	0.3269	26.96
4	85	0.12	3583.33	34.78	0.2735	27.42

In [6]:
```
df_241UDS0111
```

Out[6]: Columns 1–8 of 14

	Year	Charging_Stations	Charging_Demand_GWh	Energy_Supplied_GWh	Charging_Cost_INR	Battery_Mineral_Price_USD	Inflation_Rate
0	2010	120	0.8	0.6	18	42.0	8.1
1	2011	165	1.1	0.9	18	45.5	7.5
2	2012	230	1.6	1.3	17	48.0	6.9
3	2013	315	2.3	1.9	17	52.0	7.2
4	2014	430	3.1	2.6	16	58.5	6.4
5	2015	590	4.4	3.7	16	64.0	5.9
6	2016	810	6.2	5.2	15	71.5	4.8
7	2017	1120	8.8	7.5	15	82.0	4.3
8	2018	1580	12.5	10.7	14	95.0	3.9
9	2019	2210	17.4	14.9	14	88.0	4.6
10	2020	3050	23.8	20.1	13	105.0	6.2
11	2021	4380	33.1	28.0	13	132.0	5.5
12	2022	6210	45.7	38.8	12	118.0	6.1
13	2023	8720	62.4	53.2	12	96.0	5.2
14	2024	11950	83.6	71.0	11	84.0	4.8
15	2025	15800	109.8	93.6	11	91.0	3.7
16	2026	20500	141.5	120.4	10	102.0	4.2

Columns 9–14 of 14 · first column repeated

	Government_Incentive_Crore	EV_Users_Million	Station_Coverage_Index	Demand_Growth_Percent	Cost_to_Mineral_Price_Ratio	Incentive_per_GWh
0	25	0.02	6000.00	0.00	0.4286	31.25
1	35	0.03	5500.00	37.50	0.3956	31.82
2	48	0.05	4600.00	45.45	0.3542	30.00
3	62	0.08	3937.50	43.75	0.3269	26.96
4	85	0.12	3583.33	34.78	0.2735	27.42
5	110	0.18	3277.78	41.94	0.2500	25.00
6	145	0.27	3000.00	40.91	0.2098	23.39
7	190	0.40	2800.00	41.94	0.1829	21.59
8	250	0.60	2633.33	42.05	0.1474	20.00
9	330	0.88	2511.36	39.20	0.1591	18.97
10	420	1.25	2440.00	36.78	0.1238	17.65
11	560	1.80	2433.33	39.08	0.0985	16.92
12	720	2.55	2435.29	38.07	0.1017	15.75
13	910	3.50	2491.43	36.54	0.1250	14.58
14	1150	4.70	2542.55	33.97	0.1310	13.76
15	1420	6.10	2590.16	31.34	0.1209	12.93
16	1750	7.80	2628.21	28.87	0.0980	12.37

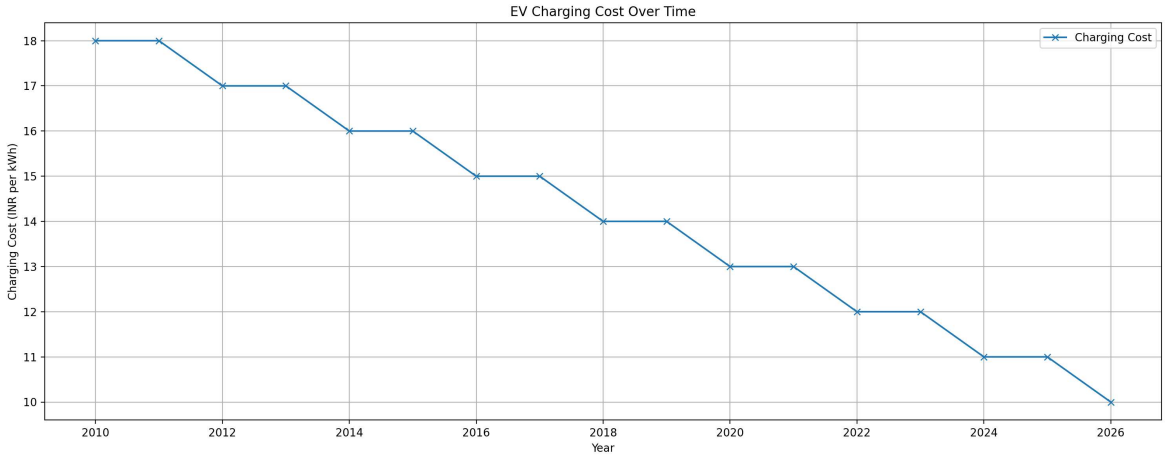
In [7]:
```
df_241UDS0111.columns
```

Out[7]:
```
Index(['Year', 'Charging_Stations', 'Charging_Demand_GWh',
       'Energy_Supplied_GWh', 'Charging_Cost_INR', 'Battery_Mineral_Price_USD',
       'Inflation_Rate', 'Government_Incentive_Crore', 'EV_Users_Million',
       'Station_Coverage_Index', 'Demand_Growth_Percent',
       'Cost_to_Mineral_Price_Ratio', 'Incentive_per_GWh'],
      dtype='str')
```

In [8]:
```
df_241UDS0111["Year"] = pd.to_numeric(df_241UDS0111["Year"])
```

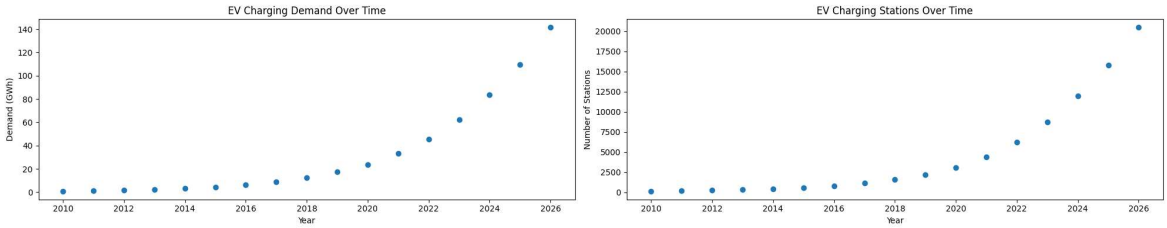
In [ ]:
```
## Step 3: Visualize Data Using Various Plots
### 1. EV Charging Cost Over Time
```

In [9]:
```
plt.figure(figsize=(15, 6), dpi=200)
plt.plot(
    df_241UDS0111["Year"],
    df_241UDS0111["Charging_Cost_INR"],
    marker="x",
    label="Charging Cost"
)
plt.title("EV Charging Cost Over Time")
plt.xlabel("Year")
plt.ylabel("Charging Cost (INR per kWh)")
plt.tight_layout()
plt.legend()
plt.grid(True)
plt.show()
```



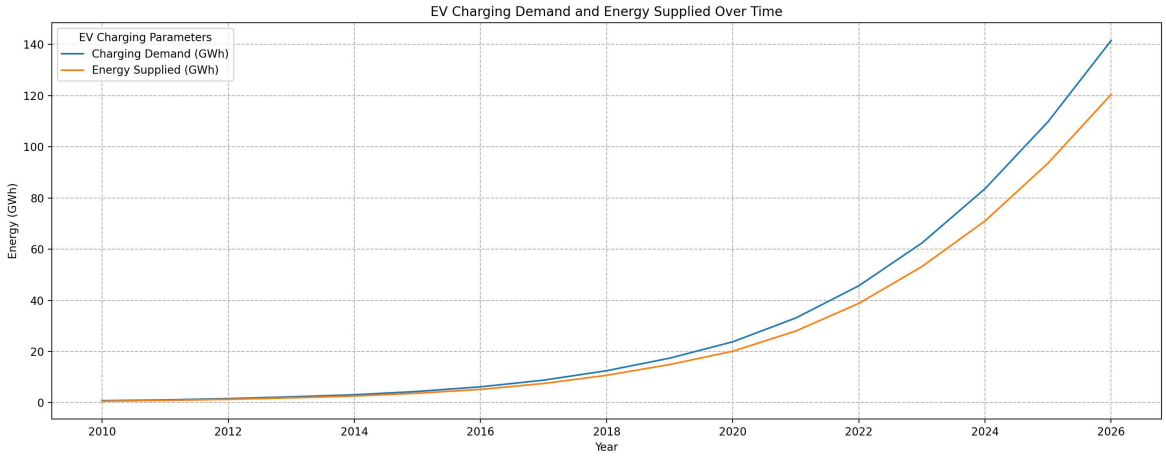
In [ ]:
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## Step 4: Subplots
```

In [10]:
```
fig, axes = plt.subplots(1, 2, figsize=(20, 4))
axes[0].scatter(df_241UDS0111["Year"], df_241UDS0111["Charging_Demand_GWh"])
axes[0].set_title("EV Charging Demand Over Time")
axes[0].set_xlabel("Year")
axes[0].set_ylabel("Demand (GWh)")
axes[1].scatter(df_241UDS0111["Year"], df_241UDS0111["Charging_Stations"])
axes[1].set_title("EV Charging Stations Over Time")
axes[1].set_xlabel("Year")
axes[1].set_ylabel("Number of Stations")
plt.tight_layout()
plt.show()
```



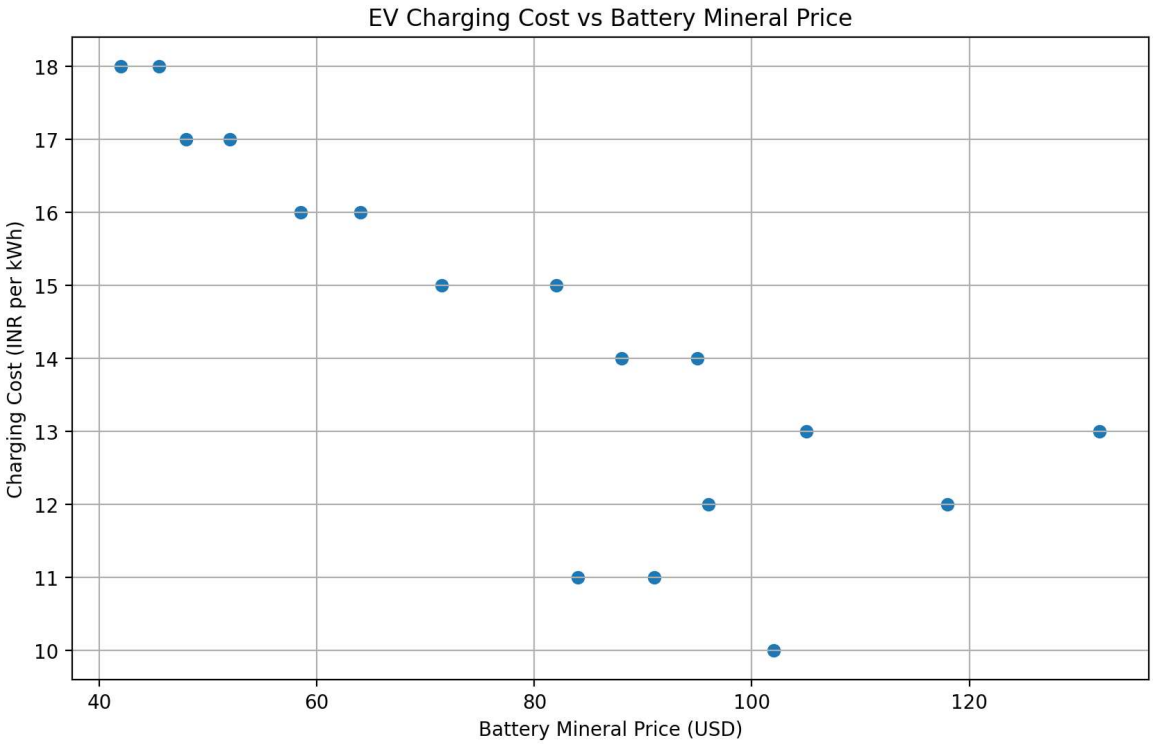
In [ ]:
```
## Step 5: Comparing Charging Demand, Energy Supplied and Charging Stations
```

In [11]:
```
plt.figure(figsize=(15, 6), dpi=200)
plt.plot(df_241UDS0111["Year"], df_241UDS0111["Charging_Demand_GWh"], label="Charging Demand (GWh)")
plt.plot(df_241UDS0111["Year"], df_241UDS0111["Energy_Supplied_GWh"], label="Energy Supplied (GWh)")
plt.title("EV Charging Demand and Energy Supplied Over Time")
plt.xlabel("Year")
plt.ylabel("Energy (GWh)")
plt.tight_layout()
plt.legend(loc="upper left", title="EV Charging Parameters")
plt.grid(True, linestyle="--")
plt.show()
```



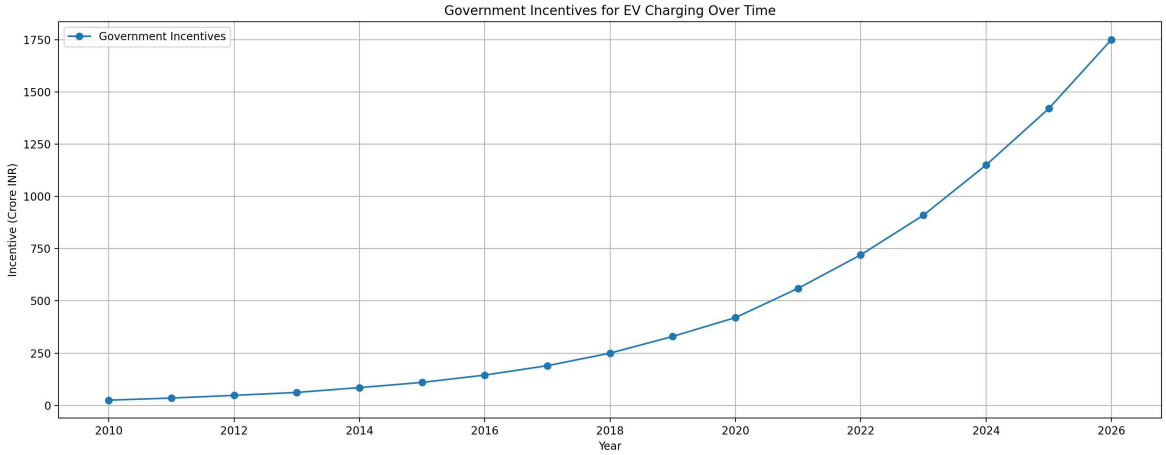
In [ ]:
```
## Step 6: Charging Cost vs Battery Mineral Price
```

In [12]:
```
plt.figure(figsize=(10, 6), dpi=200)
plt.scatter(df_241UDS0111["Battery_Mineral_Price_USD"], df_241UDS0111["Charging_Cost_INR"])
plt.title("EV Charging Cost vs Battery Mineral Price")
plt.xlabel("Battery Mineral Price (USD)")
plt.ylabel("Charging Cost (INR per kWh)")
plt.grid(True)
plt.show()
```



In [ ]:
```
## Step 7: Government Incentives Over Time
```

In [13]:
```
plt.figure(figsize=(15, 6), dpi=200)
plt.plot(df_241UDS0111["Year"], df_241UDS0111["Government_Incentive_Crore"], marker="o",
label="Government Incentives")
plt.title("Government Incentives for EV Charging Over Time")
plt.xlabel("Year")
plt.ylabel("Incentive (Crore INR)")
plt.tight_layout()
plt.legend()
plt.grid(True)
plt.show()
```



In [ ]:
```
## RESULT
Matplotlib was used to visualize EV charging costs, demand, charging infrastructure, supplied energy,
battery-mineral price relationships, and government incentive trends over time.
```